Full-Stack Laravel Web Application

Expenser

A Premium, Elegant Group Expense-Sharing & Debt Settlement Platform

FULL-STACK DEV
Lead Developer

UI/UX ARCHITECT
Design Specialist

ALGORITHM ENGINEER
Systems Engineer

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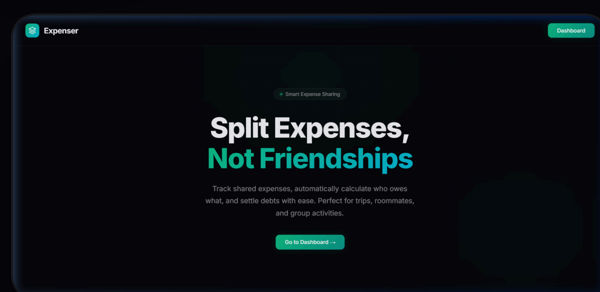
01 / INTRODUCTION

The Challenge of Group Expenses

Managing shared finances during trips, room shares, or group dinners is historically tedious and prone to friction.

- **Tracking complexity:** Who paid what, when, and for whom?
- **Flexible splits:** Not all bills are shared equally.
- **Settlement chaos:** Dozens of messy peer-to-peer transfers.
- **Expenser Solution:** A premium, real-time shared ledger that automatically calculates shares, validates balances, and minimizes transfer routes.



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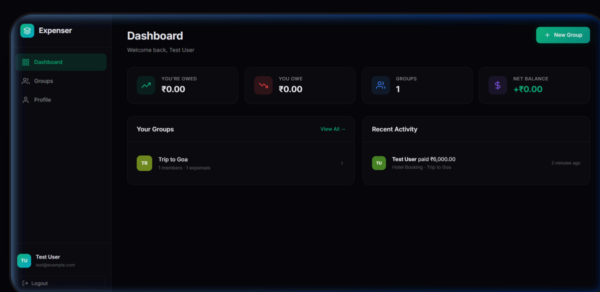
02 / SYSTEM CAPABILITIES

Rich, Flexible Feature Suite

Expenser goes beyond basic logging, incorporating dynamic user authentication, full group control, and detailed financial ledger capabilities.

- **Group Membership:** Create custom spaces (e.g. trips, flats) and add users by email in real-time.
- **Dynamic Splits:** Split equally, by exact amount, or by percentage.
- **Smart Ledger:** Detailed, audited log of every expense and settlement payment.
- **Direct Settle:** Integrated system to register peer-to-peer payments and clear balances in a single click.



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03 / TECH STACK

Engineered for Stability & Speed

Leveraging modern web architecture to deliver a robust, highly responsive, and clean development ecosystem.

L

Laravel 12 Framework

Eloquent ORM, routing, middleware, Breeze Authentication scaffold.

S

SQLite / DB Engine

Fast relational database with structured tables & integrity constraints.

V

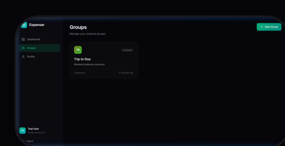
Vite Asset Pipeline

Extremely fast hot-reload asset bundling and build system.

C

Vanilla CSS3 & JS

Bespoke dark theme, responsive grids, custom CSS variables, zero heavy frameworks.



Project Presentation

Slide 4

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04 / SYSTEM ARCHITECTURE

Relational Database & Models Layout

Clean database normalization prevents redundancy and ensures strict data integrity between members, transactions, shares, and groups.

- **Users & Groups:** Many-to-many relationship tracked via `group_user` pivot table.
- **Expenses:** Belongs to a single group and is paid by one user.
- **Expense Shares:** Keeps individual share amounts per user per expense, allowing flexible split schemes.
- **Settlements:** Links two users in a group to record peer-to-peer balance offsets.

DATABASE SCHEMAS

Groups: id, name, description, created_by, timestamps

Group_User: group_id, user_id, joined_at

Expenses: id, group_id, paid_by, description, amount, split_type

Expense_Shares: id, expense_id, user_id, share_amount

Settlements: id, group_id, paid_by, paid_to, amount, note

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05 / CORE ENGINE

Real-Time Balance Calculation

The system computes live balances for all members dynamically, preventing desync errors and keeping accurate state.

- **Credit Step:** Group expense payer gets credited `+Amount`.
- **Debit Step:** Every sharing user gets debited `-Share_Amount`.
- **Settlement Correction:** Peer settlements decrease the payer's balance and increase the receiver's balance.
- **Net Formula:** `Balance = Paid_Expenses - Shared_Cost - Sent_Payments + Received_Payments`

```
// Balance calculation kernel
foreach ($expenses as $expense) {
    $balances[$expense->paid_by] += $expense->amount;
    foreach ($expense->shares as $share) {
        $balances[$share->user_id] -= $share->share_amount;
    }
}
foreach ($settlements as $settlement) {
    $balances[$settlement->paid_by] -= $settlement->amount;
    $balances[$settlement->paid_to] += $settlement->amount;
}
```

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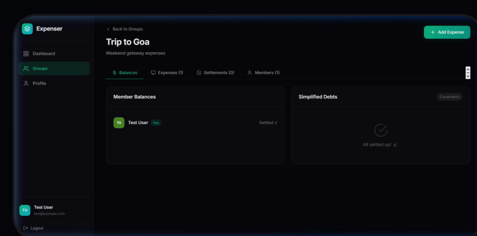
Min-Cash-Flow Debt Simplification

Expenser implements a greedy algorithm to reduce peer-to-peer transfers, resolving group debts in the fewest transactions possible.

- **Algorithm Class:** Greedy optimization (Min Cash Flow).
- **State Separation:** Members divided into Net Creditors and Net Debtors.
- **Greedy Step:** Matches the largest creditor with the largest debtor and schedules a direct payment.
- **Effect:** Reduces a group of N members from potentially N*(N-1) transfers down to a maximum of N-1 transfers.



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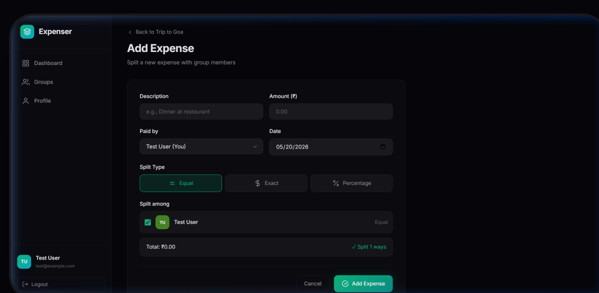
07 / INTERACTIVE UI

Dynamic & Smart Expense Logging

Built with custom JavaScript to validate splits interactively, giving immediate feedback to prevent math errors before submission.

- **Live Calculation:** Displays split outcomes immediately as values change.
- **Validation:** Exact amounts must sum to the total; percentages must equal exactly 100%.
- **Error Handling:** Warns users if the values are unbalanced before submission.
- **Reminders:** Rounding remainders are handled by allocating to the primary payer.



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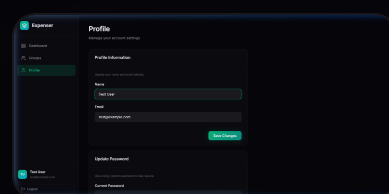
08 / UI AESTHETICS

Custom Dark Theme & Glassmorphism

Crafted from the ground up without UI frameworks to ensure a highly responsive, modern, and lightweight design system.

- **Color System:** HSL curated deep dark background tones mixed with emerald and teal primary colors.
- **Backdrop Filters:** Frosted glass transparency overlays with fine borders.
- **Fluid Animations:** GPU-accelerated translate hover states and keyframe fade-in transitions.
- **Responsive Flex:** Responsive sidebars and grids that adjust to all viewports.



09 / Conclusion & Future

Expenser Project Summary

Successfully created a fast, beautiful, and computationally efficient expense sharing system that minimizes financial friction and transaction routes in group settings.

Accomplishments

✓ Modern dark-themed user interfaces

✓ 3 separate mathematical split structures

✓ Greedy debt minimizer engine

✓ Full responsiveness and auth flows

Future Vision

• QR-code invitation links

• Multi-currency support + exchange API

• Push notifications & email summaries

• Native mobile application wrap